

School Bus Temperature Monitor – Dual Screen

Operational Summary

To begin, simply plug in the USB plug into a USB port. This device monitors up to two DS18B20 temperature sensors (one for inside the bus, one for outside) and shows the readings on two small LCD screens.

It may also be plugged into a computer or system that allows monitoring over USB Serial ports to receive or log output data for scientific evaluation.

The LED lights and LCD backlights will turn off and go into Economy Mode after about 1 hour in Turbo Mode and about 2 hours in Default Mode.

Hardware

- **DS18B20 sensors:** Both connected in parallel to pin D2 for Data.
 - 4.7K Ohm resistor between 5V and Data line.
- **LCD screens:** Two I2C LCDs (one at address 0x27, one at 0x23) – done by bridging A2 pins together on the second LCD screen. Both I2C connectors are connected to: A4 -> SDA, A5 -> SCL. They are LCD1602 units. An LCD2004 could work but the code would need to be modified for a 20x4 screen as opposed to dual 16x2 screens.
- **Lights:** Headlights on D9 & D10, Top lights on D11 & D12. 5V pre-wired lights with resistors included.
- **Reset button:** Press to restart the system and turn the LCD backlights and LED lights back on, connected to the RST pin on the Arduino Nano. Adafruit 16mm Panel Mount Momentary Pushbutton - Red (product #1445).
- **Turbo Mode Switch:** Under the hood. Enable to decrease temp resolution to 10 from the default of 12 and decrease wait times in sending data to across the serial wire. Default Mode loops readings every 3.35 seconds (39 loops per minute) where Turbo Mode loops readings every 1.55 seconds (18 loops per minute). Economy Mode is activated in about 111 minutes in Default Mode and about 52 minutes in Turbo Mode. A “!” will appear in the bottom-right of the screen when Turbo Mode is active. It is safe to leave the unit in Turbo Mode as long as you wish, but readings will be rounded to 0.5 degree steps. Adafruit Mini Panel Mount SPDT Toggle Switch (product #3221). Most low-voltage switches would work as a replacement.
- **Paint** - Behr Blazing Bonfire Satin, Behr Cracked Pepper Satin

Operation & Errors

- **Devices: 1** or **Devices: 2** → A sensor is missing.
- **ERROR** or **0 Sensors Found** → Multiple sensors are missing.
- **Too Many Sensors** → More than 2 sensors detected (only 2 are supported).
- **Sensor X Missing** → One sensor is not responding.
- **No Display**: Check power and I2C wiring for the LCD screens.
- **-196.60** or **196.60 showing**: Bad sensor reading – check connections and 4.7K resistor.
- **Lights won't come on**: Press Reset button. Standard LEDs with a low-value resistor could be used as replacements and reconnected if they die. Clip the existing wires and rejoin them to the new wires or resolder the connectors onto the circuit board.
- **Flicker or Error Behavior**:
 - Touching Connectors - A connector could be touching inside the bus, causing a short-circuit. Open it up using a screwdriver and rearrange the wires.
 - Insufficient Power - Unlikely. A low-power USB port may also cause errors. The unit does function properly on USB 2.0 ports, which only provide 2.5W.
 - **Opening the Hood**: Carefully squeeze the hood (try not to damage the paint) and pull upwards. Additionally, an allen wrench can be placed into the “engine compartment” from a hole in the “passenger compartment” to pop the hood open.

Temperature Resolution <DallasTemperature.h> Software Library

- Turbo Mode sets Resolution at 10 bits, effectively stepping at ~0.5F
- Default Mode sets Resolution at 12 bits, effective stepping at ~0.1F

Resolution (bits)	Smallest Step in °C	Smallest Step in °F	Max Conversion Time	Typical Use Case
9	0.5°C	0.9°F	~93.75 ms	Fastest reads, lower precision
10	0.25°C	0.45°F	~187.5 ms	Good balance
11	0.125°C	0.225°F	~375 ms	Higher precision
12 (default)	0.0625°C	0.1125°F	~750 ms	Highest precision, slowest

Serial Monitoring with Arduino IDE

Upon startup, the device communicates over USB at 9600 Baud – slow, but highly reliable if USB extension cables are used. It could be paired with a data logger, such as one written in Python that creates CSV files. However, Arduino IDE comes with a built-in Serial Monitor and Serial Plotter to display a graph. Startup of the unit takes under 10 seconds typically.

The Temp_CPU variable is also output to serial and is a rough estimate of the CPU temperature on the Arduino Nano. It could be interesting to observe how it is warmer than the air temperature, especially in a cold environment. It would be useful to show if the CPU is overheating, but that is a rare condition but may be present as the unit ages.

Arduino IDE Serial Monitor & Plotter Settings - Display Messages

- Select a COM Port: Tools -> Port
 - Try available ports until the correct one is found. COM3 and COM4 are typical.
- Tools -> Serial Monitor or Tools -> Serial Plotter
- Autoscroll, Newline, 9600 Baud
- Several online videos of operation are available as are different serial monitor or plotter options for live graphing.
- A history of the serial plot could be copied-pasted into a CSV, or into an online input and turned into a graph.



